

Lecture 4

CMOS Inverter & Layout Design

Prof Peter YK Cheung

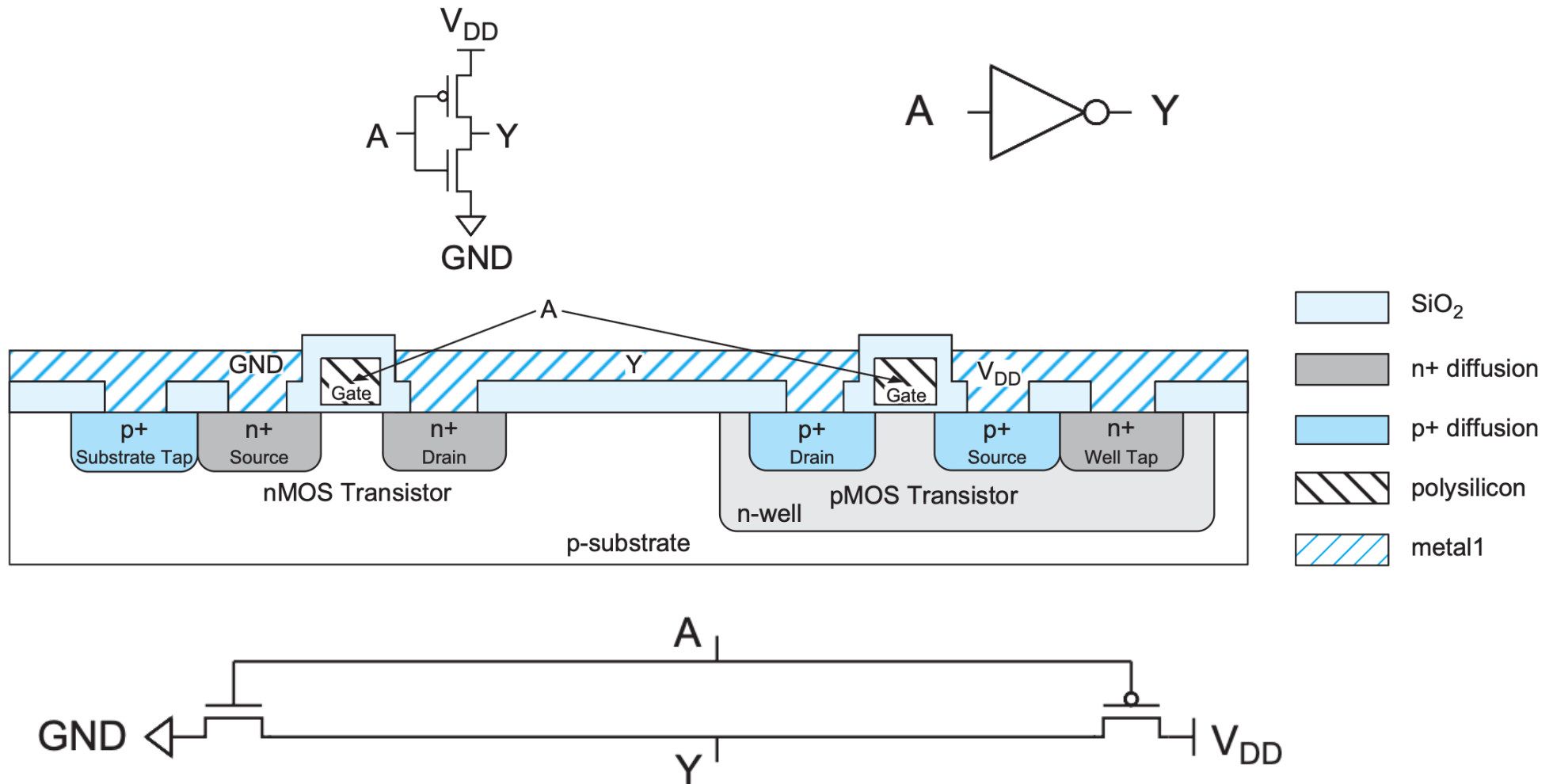
Department of Electrical & Electronic Engineering



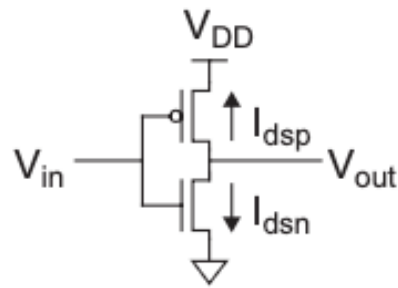
URL: <https://elec70142-digital-vlsi-design.github.io/course-website/>

E-mail: p.cheung@imperial.ac.uk

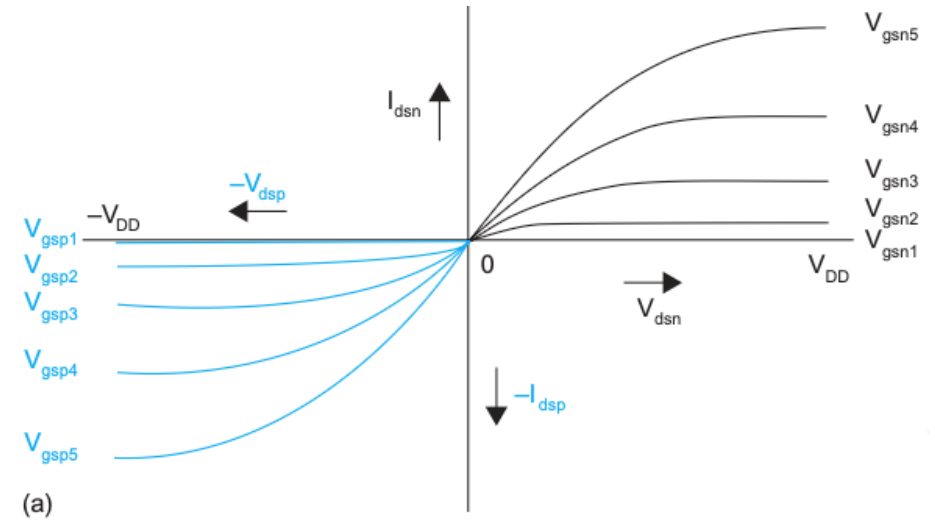
Cross section of an Inverter



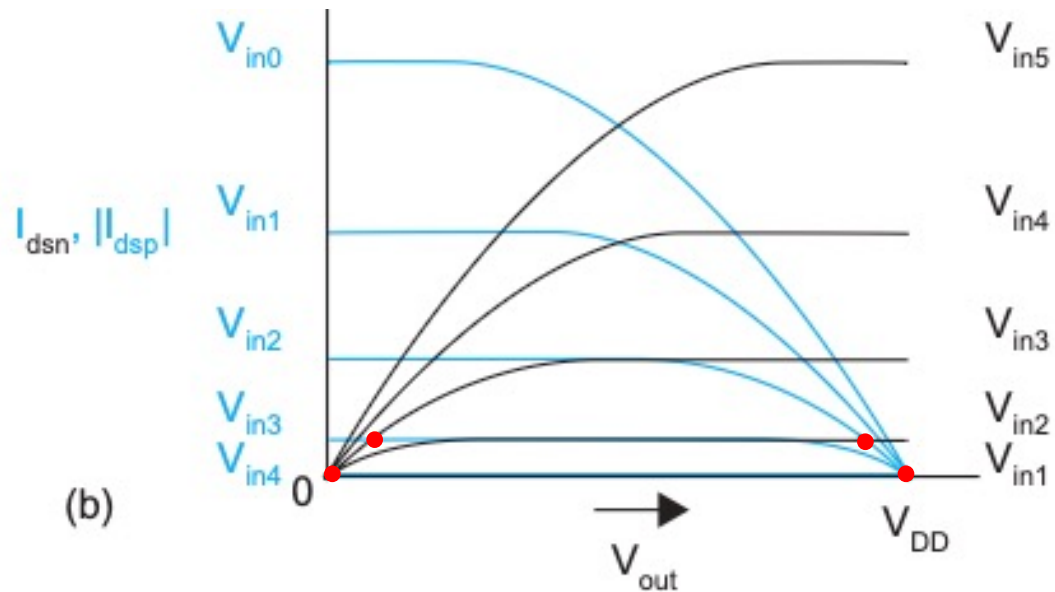
Inverter DC characteristics



$$I_{dsn} = |I_{dsp}|$$

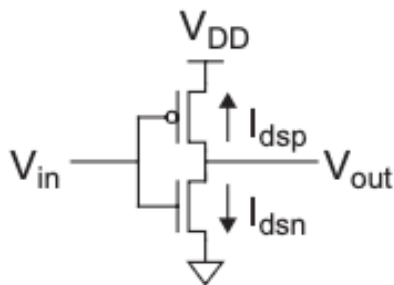


$$I_{dsn} = |I_{dsp}|$$

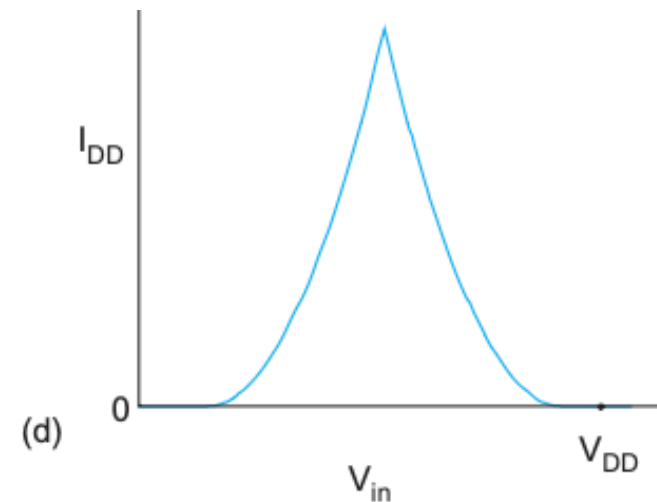
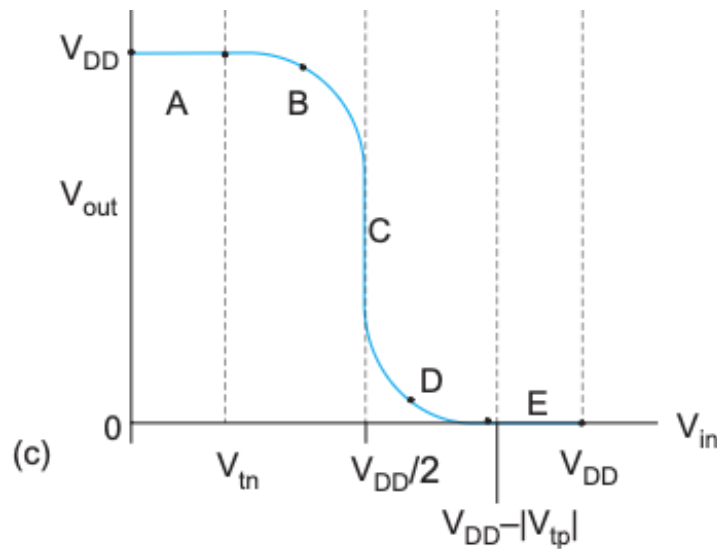


W&H p87-88

Inverter switching threshold



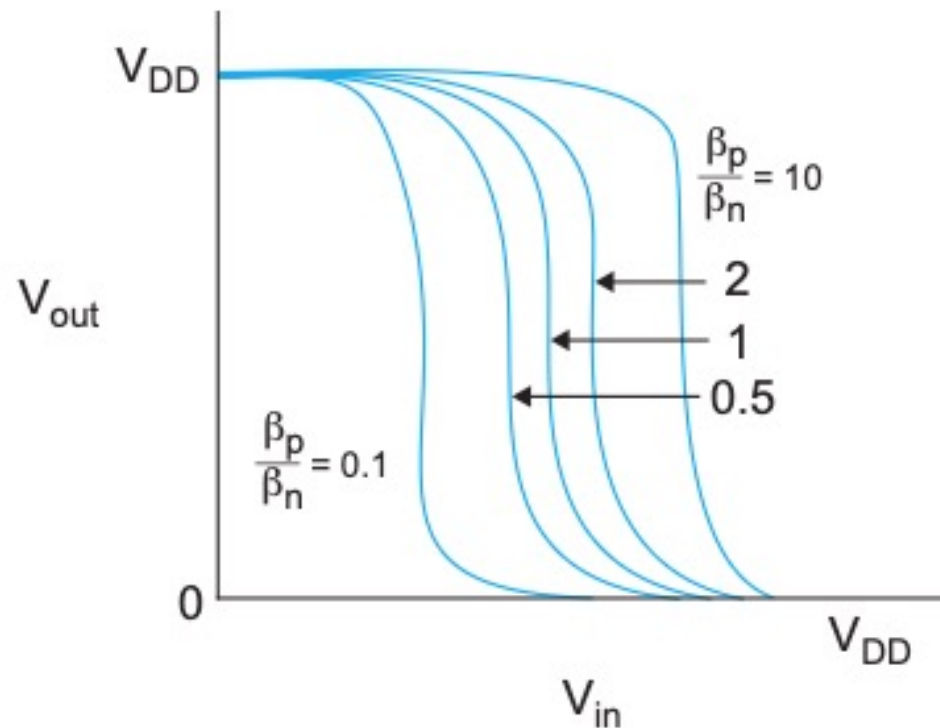
Region	Condition	p-device	n-device	Output
A	$0 \leq V_{in} < V_{tn}$	linear	cutoff	$V_{out} = V_{DD}$
B	$V_{tn} \leq V_{in} < V_{DD}/2$	linear	saturated	$V_{out} > V_{DD}/2$
C	$V_{in} = V_{DD}/2$	saturated	saturated	V_{out} drops sharply
D	$V_{DD}/2 < V_{in} \leq V_{DD} - V_{tp} $	saturated	linear	$V_{out} < V_{DD}/2$
E	$V_{in} > V_{DD} - V_{tp} $	cutoff	linear	$V_{out} = 0$



W&H p88-89

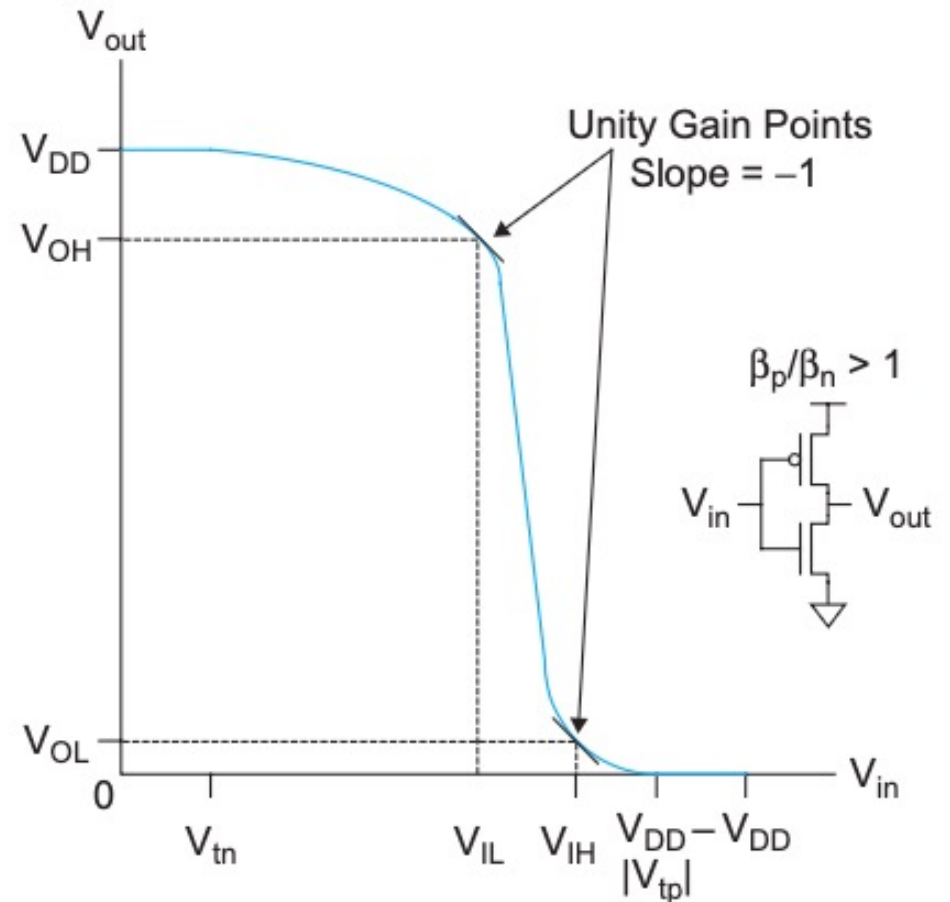
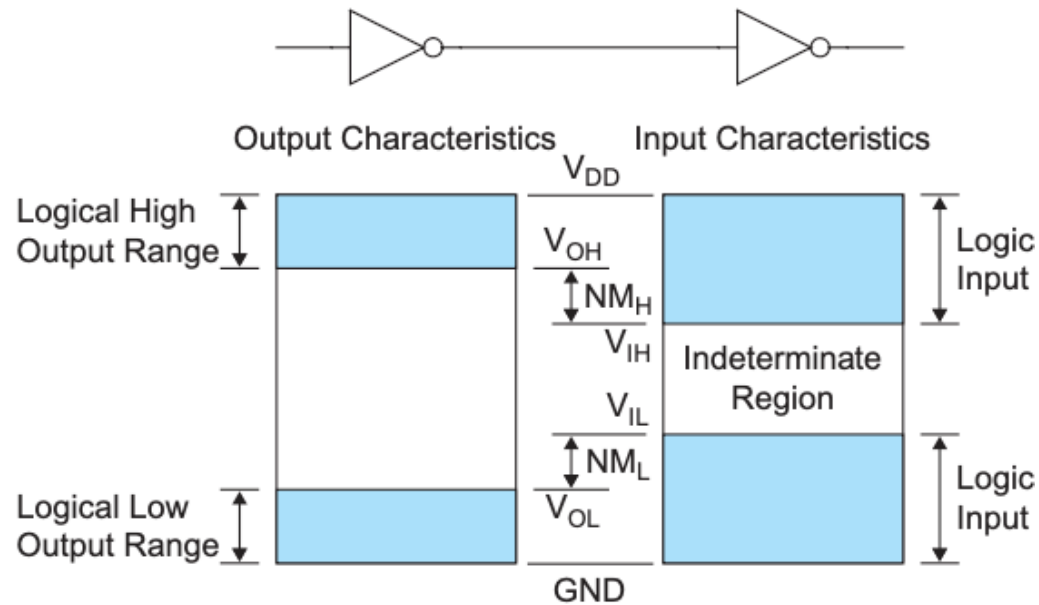
Beta ratio vs switching threshold

- ◆ Extract switching point depends on β_p / β_n
- ◆ If $\beta_p / \beta_n = 1$, switching occurs at around $V_{DD}/2$



W&H p90

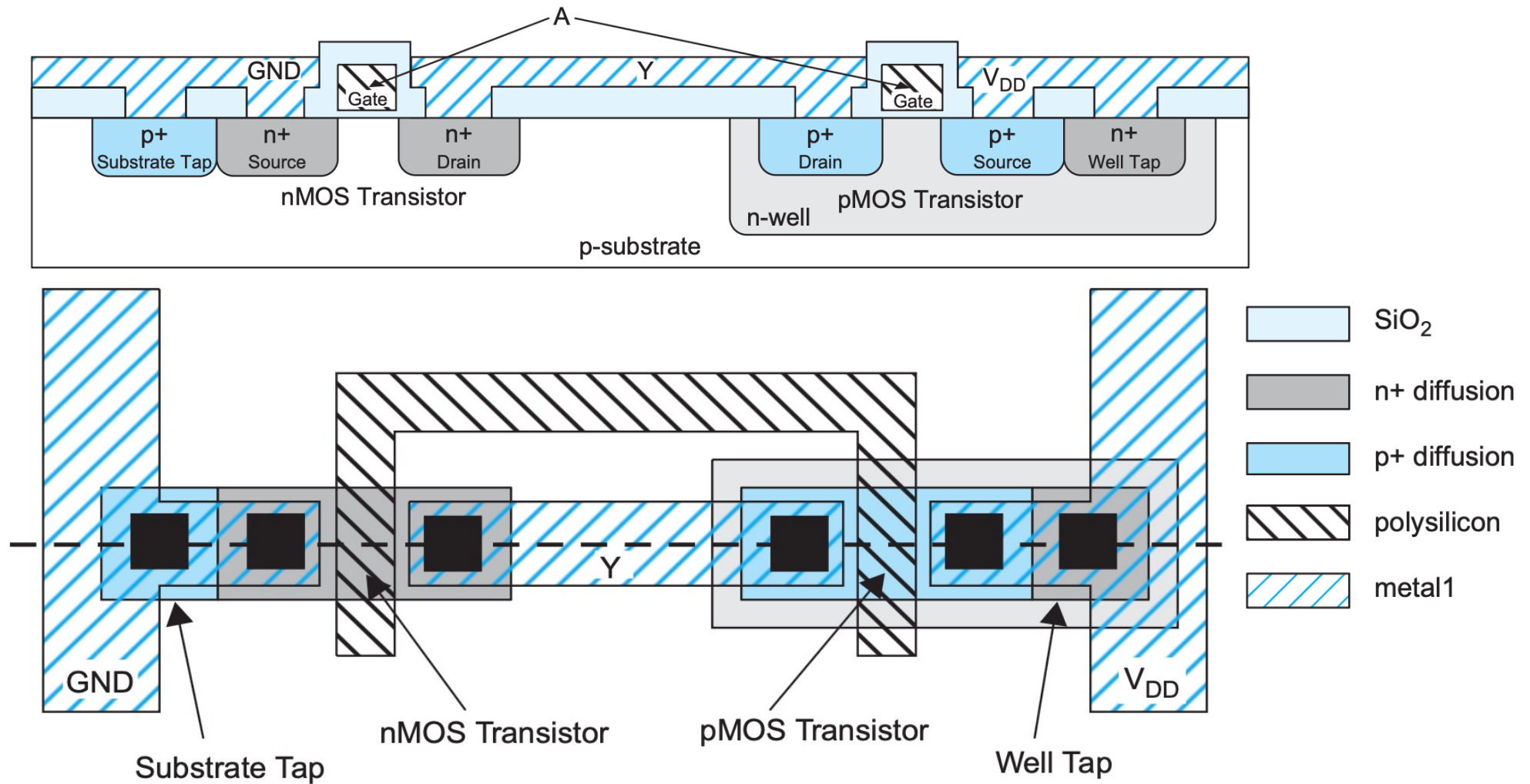
Noise Margin in digital circuits



- ◆ Select logic levels at unity gain point of DC transfer characteristic

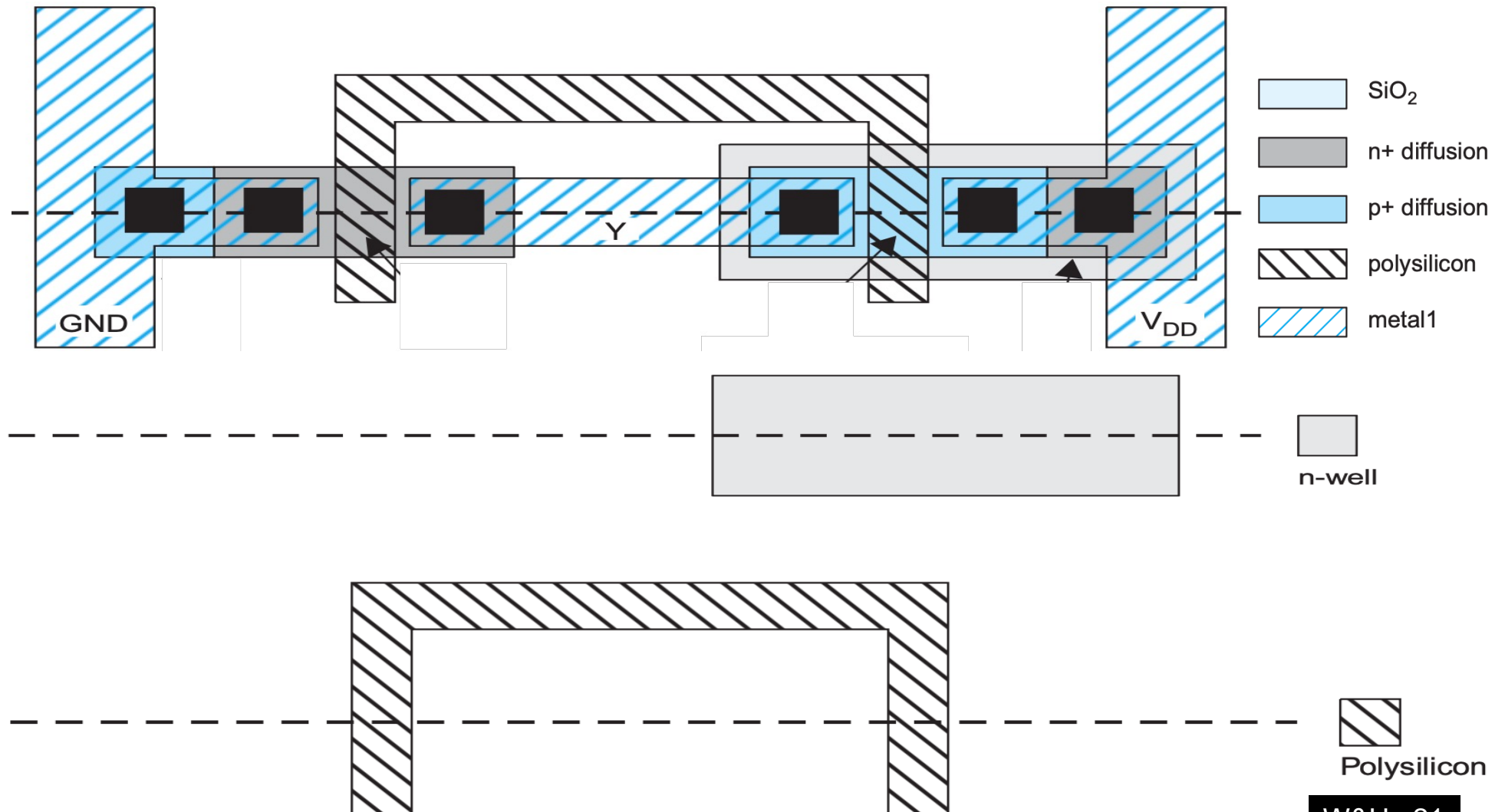
W&H p91-92

Cross-section & Layout of Inverter

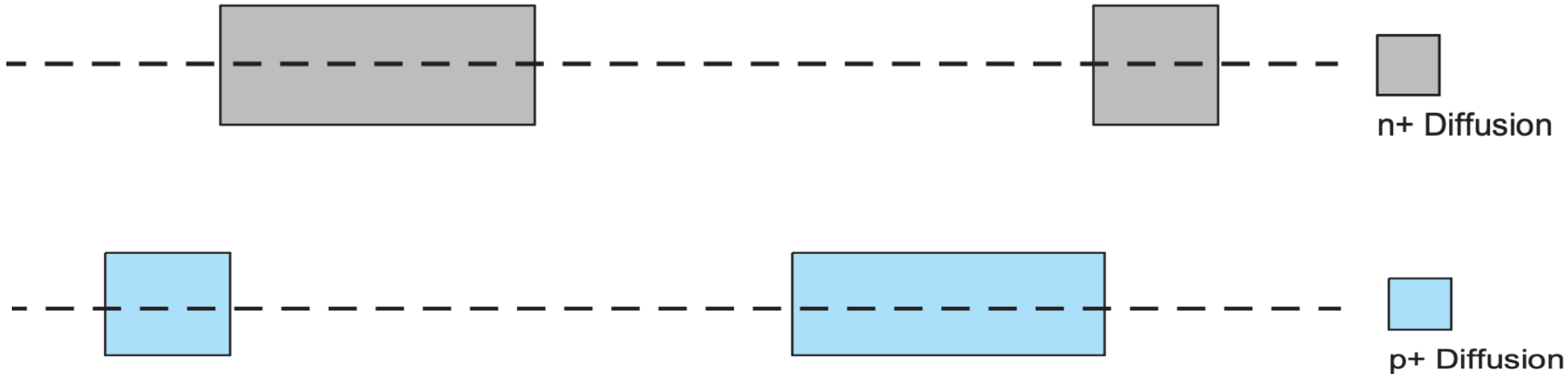


W&H p21

Mask layout design of an Inverter (1)

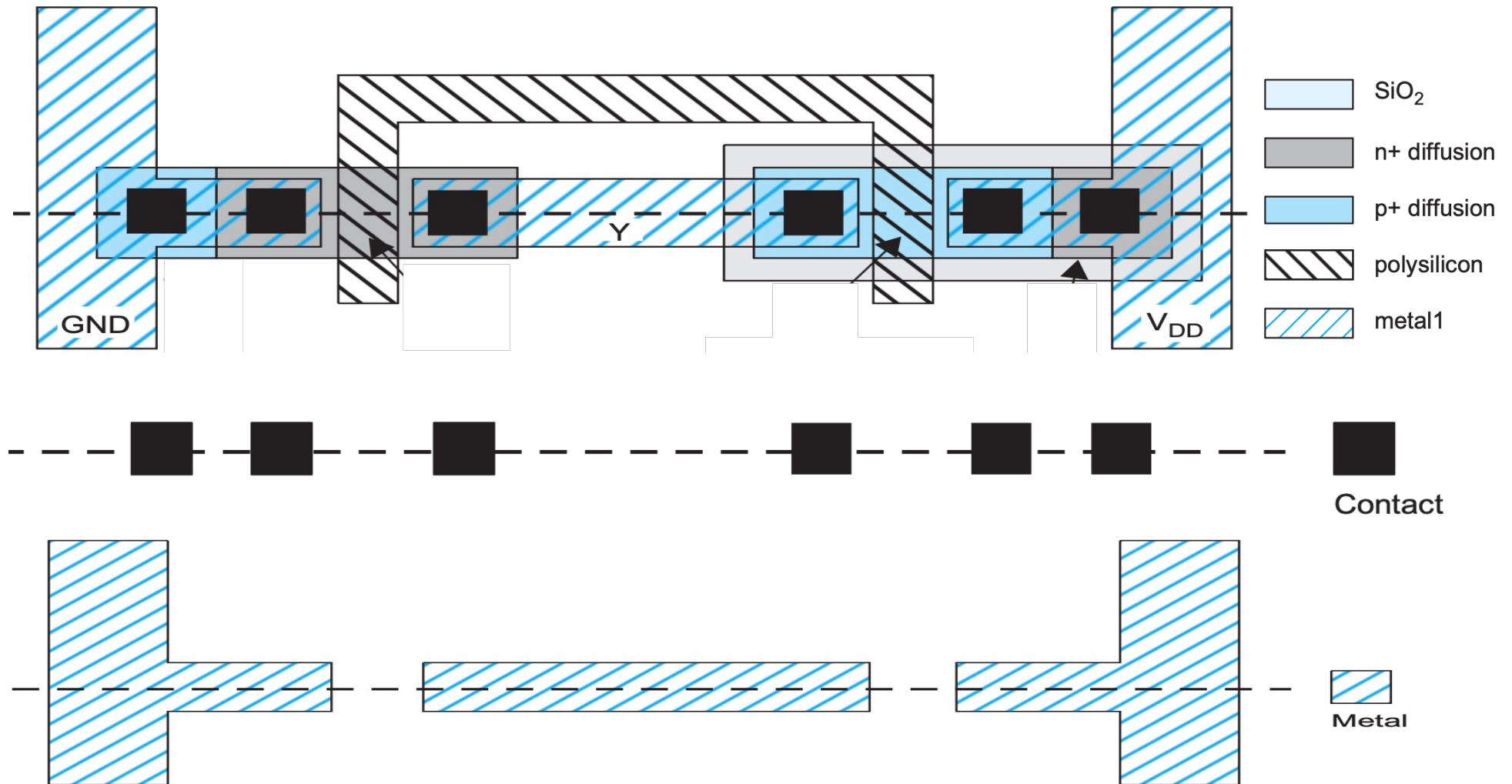


W&H p21

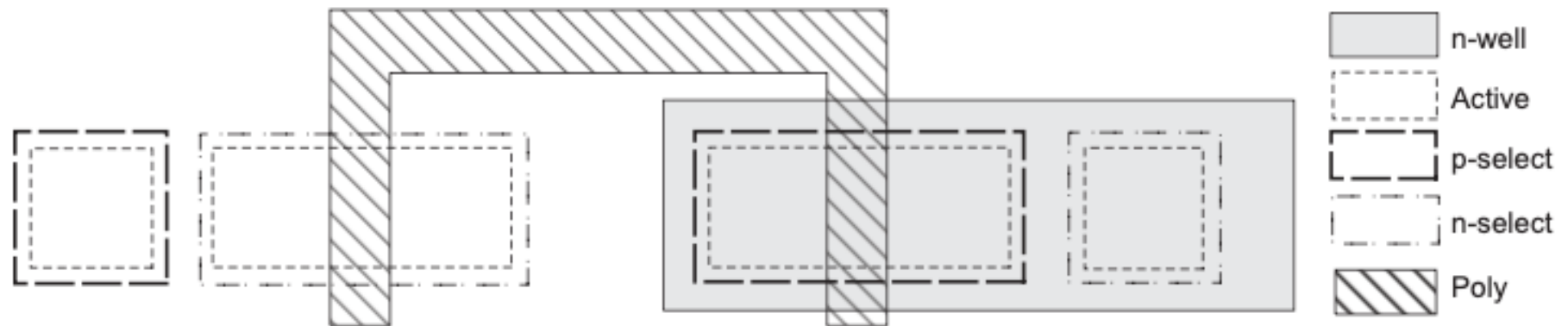


W&H p21

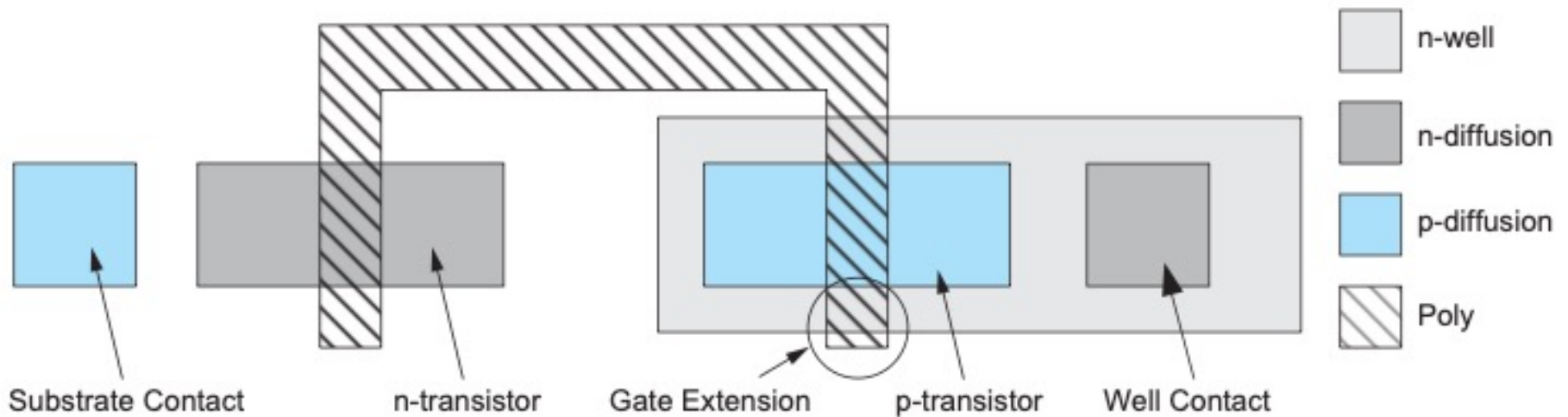
Mask layout design of an Inverter (3)



Mask set vs Fabricated circuit



(a)



Substrate Contact

n-transistor

Gate Extension

p-transistor

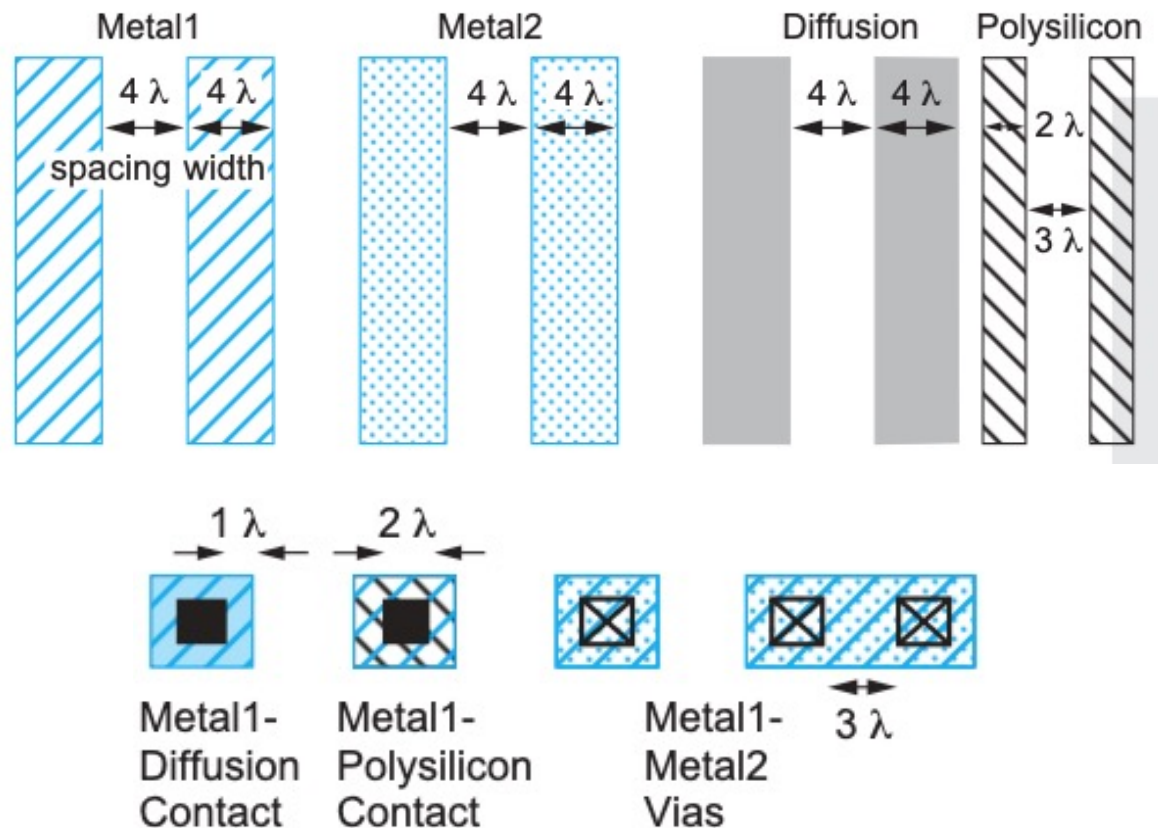
Well Contact

(b)

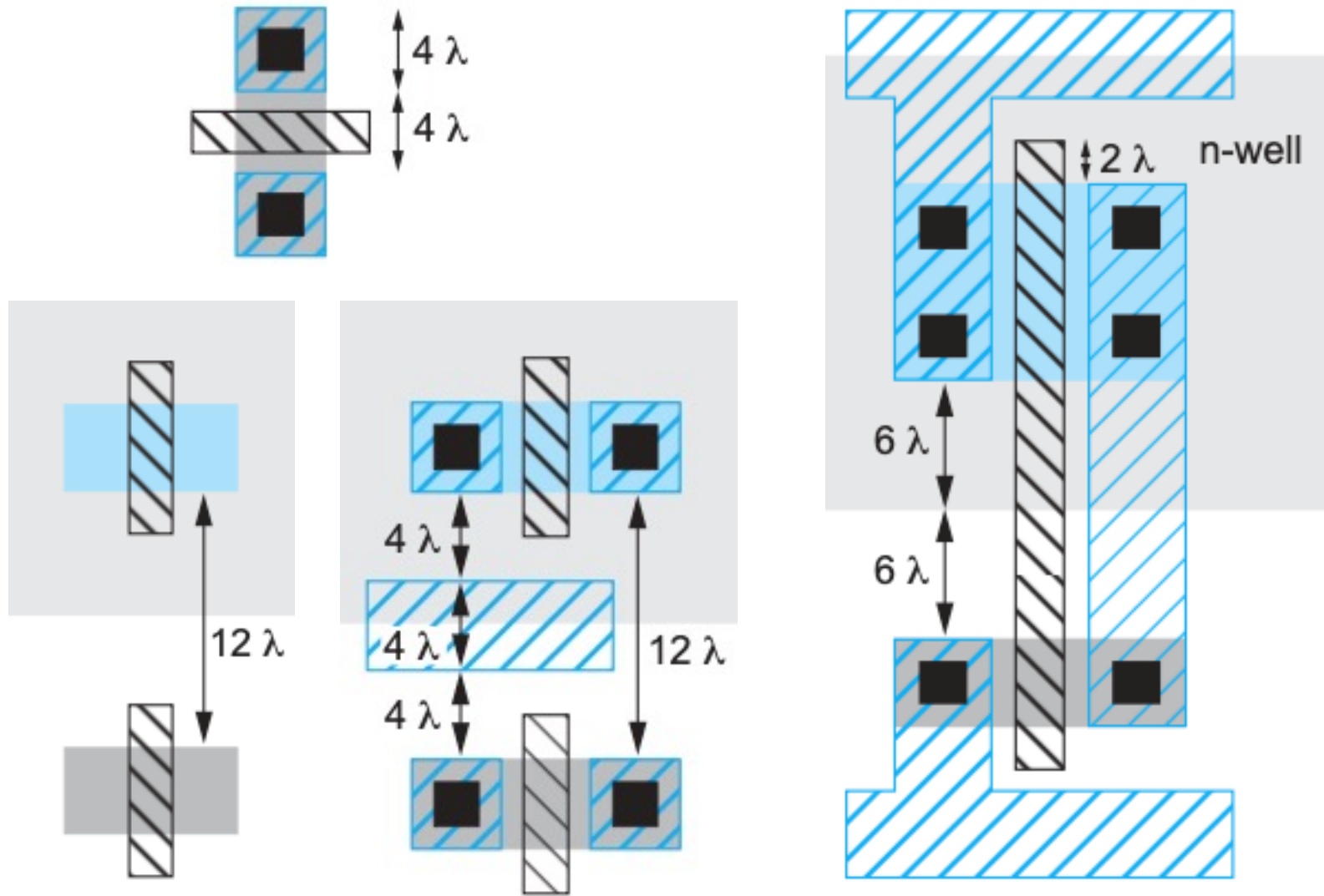
W&H p155

Scalable Layout Design Rules (1)

- ◆ Design Rules specify geometric constraints on the layout to guarantee functioning chips.
- ◆ Lambda λ is unit of measure often used to define the rules.
- ◆ Unusually half of the smallest feature size (e.g. for 180nm, $\lambda = 90\text{nm}$)



Scalable Layout Design Rules (2)



Absolute Layout Design Rules

Layer / Feature	Rule	Typical Value (μm)	Notes
Poly (Gate)	Minimum width	0.07	Smallest gate length (L_{min})
Poly (Gate)	Minimum spacing	0.10	Between poly lines
Poly over Active	Extension	0.05	Poly overlap over diffusion edge
Active (Diffusion)	Minimum width	0.10	Smallest diffusion region
Active (Diffusion)	Minimum spacing	0.10	Between active regions
Contact (CA)	Contact size	0.08×0.08	Via/contact hole
Contact (CA)	Contact spacing	0.08	Edge-to-edge spacing
Metal 1	Minimum width	0.09	Local interconnect routing
Metal 1	Minimum spacing	0.09	Adjacent M1 spacing
Metal 2–6	Minimum width/spacing	0.10–0.14	Intermediate routing layers
Metal 7–9	Minimum width/spacing	0.28–0.50	Top-level and power routing
Via (V1–V8)	Size	0.08×0.08	Between metal layers
Via (V1–V8)	Spacing	0.08	Edge-to-edge spacing
N-Well / P-Well	Width / Spacing	0.40 / 0.40	For device isolation

Some Layout Guidelines

MOSFET transistors

- Keep nMOS devices grouped together near VSS (can share P+ contacts)
- Keep pMOS devices grouped together near VDD (can share N+ contacts)
- Can place multiple PMOS devices within a single N-Well (if $V_B = V_S = V_{DD}$)

Interconnections (wires)

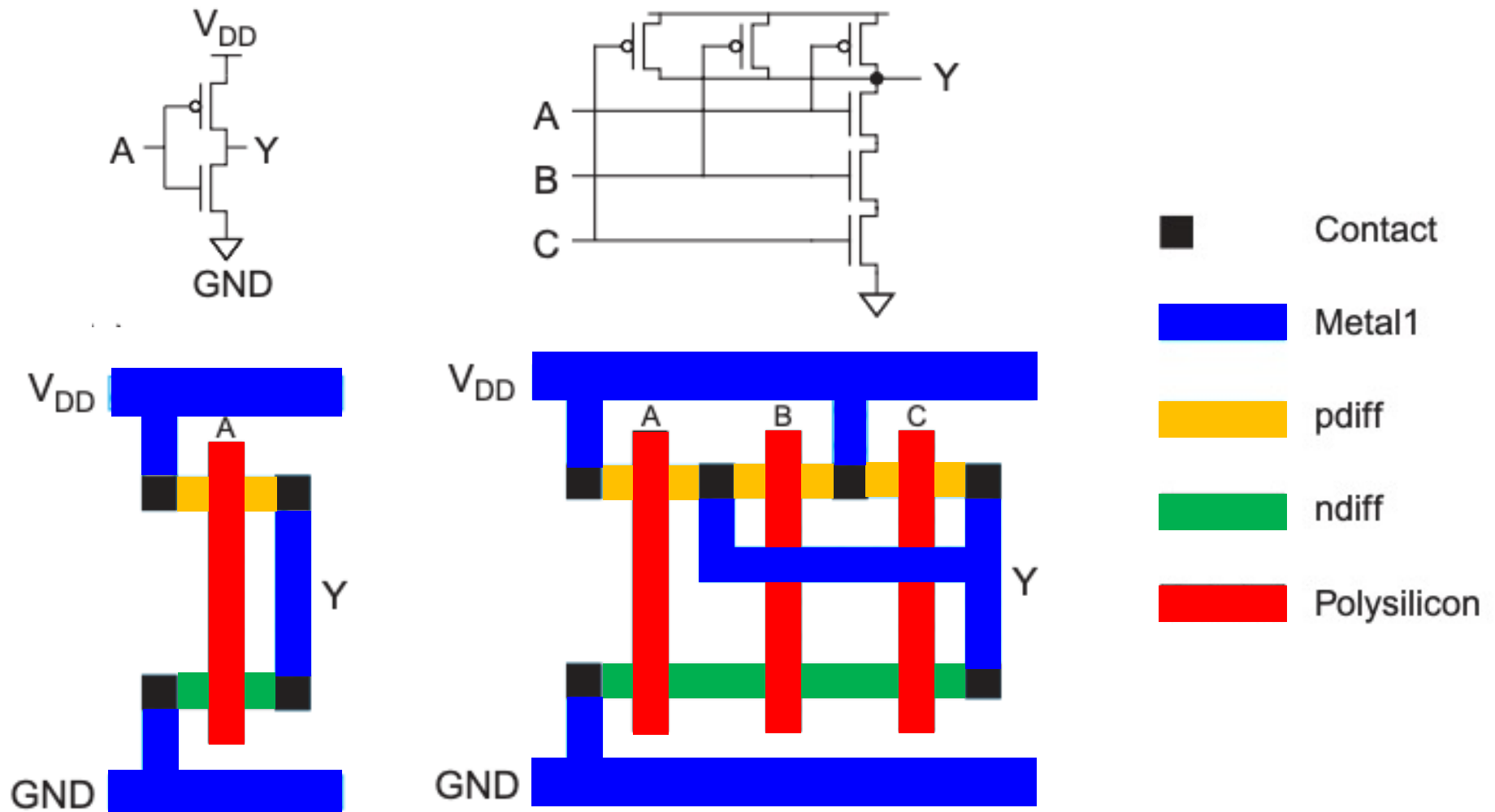
- Place pins (I/O connections) around perimeter of cell
- All long wires should use metal interconnections
- Route VDD and VSS in thick metal
- Interleave metal layers – i.e. M1/M3/M5 horizontal and M2/M4/M6 vertical
- Keep polysilicon/diffusion connections as short as possible (high resistivity)

General

- Try to design/layout as regular as possible
- Plan your layout before hand

Source: Tim Constandinou

Stick Diagram for Layout Design

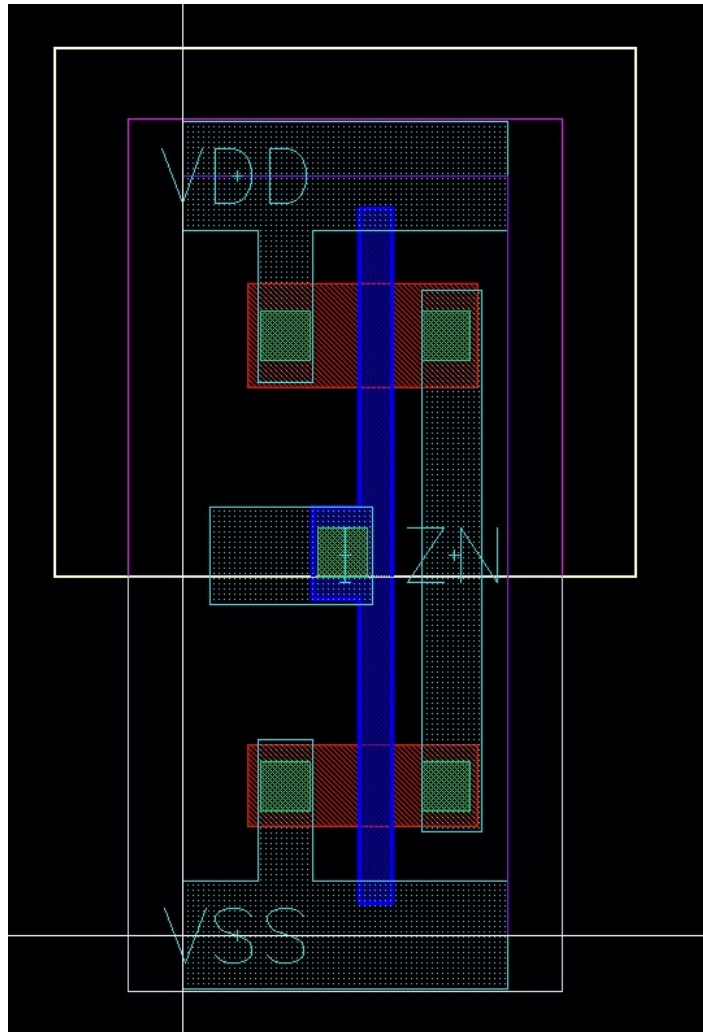


Source: Tim Constandinou

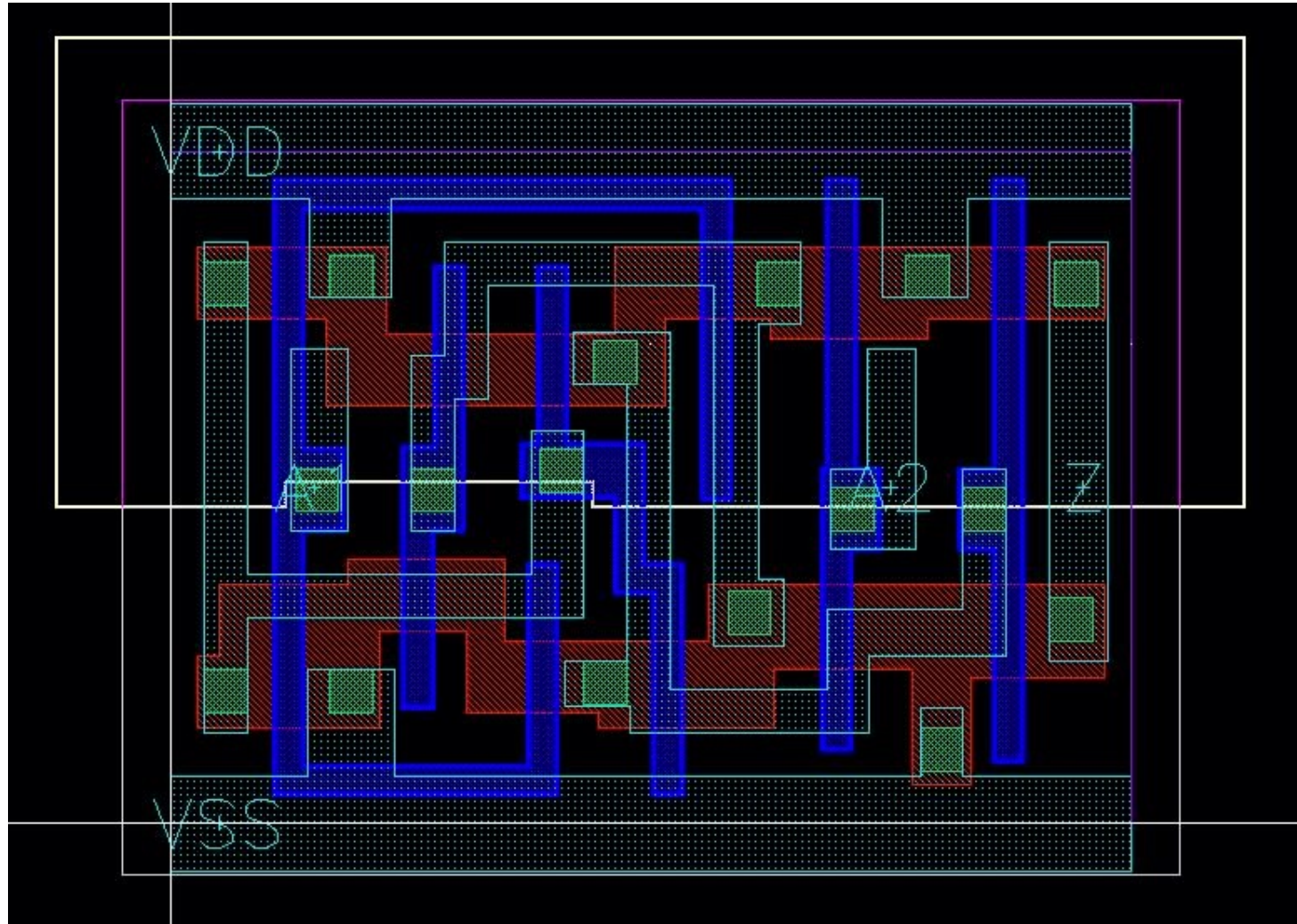
What to expect from Lab 2?

- ◆ Understand the different mask layers in a layout.
 - ◆ Read layout and “see” the circuit it implements.
 - ◆ Learn how to use the layout editor to wire things up.
 - ◆ Learn how to make sure that your design passes design rules (DRC) and the layout is the same as the schematic (LVS).
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- ◆ **Task 1:** Explore the layout of the inverter standard cell layer by layer.
 - ◆ **Task 2:** Extract transistor level circuit from XOR gate layout.
 - ◆ **Task 3:** Import circuit from Lab 1 and perform manual placement.
 - ◆ **Task 4:** Hand route the clock, control and data signals.
 - ◆ **Task 5:** Perform LVS verification.

Task 1 – Understand layout of inverter



Task 2 – Extract circuit for XOR gate



Task 3 & 4 – Floorplan, Placement and Routing

